

Nabidul Islam

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EDUCATION

City College of New York

GPA: 3.40/4.0

Bachelor of Engineering in Mechanical Engineering

Expected December 2026

- Honors: Dean's List – Spring 2023, Spring 2024 | Pearl Tsung Memorial Award – Spring 2025
- Leadership: Vice President, Tau Beta Pi (Engineering Honor Society)
- Relevant Coursework: Mechanical System Design, Mechanics of Materials, Fluid Mechanics, Thermodynamics, Heat Transfer, Dynamics, Manufacturing Processes, Computer Aided Drafting

SKILLS

CAD / FEA / CFD: SolidWorks, SolidWorks Simulation (FEA), ANSYS Fluent (CFD), ANSYS Workbench, Fusion 360

Engineering Skills: GD&T, Stress Analysis, Fatigue Analysis, Structural Analysis, 3D Printing, Soldering

Programming: MATLAB, Python, Arduino

Certifications: OSHA Laboratory Safety Standards (29 CFR 1910.1450), OSHA 10-Hour Safety Training

WORK EXPERIENCE

CarbonCLAIR

Harlem, NY

Research and Development Intern

June 2025 – Present

- Led hands-on mechanical design and prototyping for a carbon-capture device, manufacturing and assembling test hardware through iterative design cycles that achieved 60% capture efficiency.
- Executed CFD simulations in ANSYS Fluent across varying nozzle geometries and particle sizes; evaluated pressure drop, velocity profiles, and recirculation zones to identify the optimal nozzle configuration.
- Designed a structural support stand in SolidWorks and validated via FEA in SolidWorks Simulation (~10,000 elements, oak wood); confirmed factor of safety of 2.0 under 35 lb static load with no excessive deflection.
- Optimized internal airflow paths via CFD-informed geometric modifications, reducing turbulence intensity and improving particulate capture efficiency.
- Designed a hinged dust-capture mechanism in SolidWorks that expanded effective collection area by 25%, improving particulate retention during testing and deployment.

City College of New York

Harlem, NY

Course Administrator

October 2022 – March 2023

- Provided real-time technical support for audio-visual, Zoom, and OWL systems, maintaining uninterrupted operation for 99% of lectures and meetings; maintained and inspected all classroom equipment to minimize failures.

PROJECTS

GripWell | Arduino, SolidWorks, Servo Control, Closed-Loop Feedback

January 2025 – May 2025

- Led a 5-member engineering team in developing a self-stabilizing handheld tool that reduced hand tremors by up to 70% during functional testing for users with motor impairments.
- Designed and implemented a closed-loop control system using Arduino and servo actuators, using real-time sensor input to actively counteract detected tremor motion and stabilize the tool tip.
- Modeled and optimized mechanical housing components in SolidWorks, ensuring manufacturability and durability under repeated actuation cycles; validated reliability through iterative testing.

Bicycle Structural Analysis | SolidWorks, SolidWorks Simulation, FEA, Fatigue Analysis

Fall 2025

- Developed a full-assembly CAD model of a bicycle in SolidWorks, modeling all structural components (frame, fork, handlebar, seat post) with accurate geometry and material assignments.
- Performed static FEA in SolidWorks Simulation, calculating von Mises stress distributions, principal stresses, and factors of safety under representative rider loading conditions.
- Conducted fatigue analysis under cyclic loading; selected materials based on yield strength and endurance limit, and identified high-stress regions to propose geometry modifications improving structural integrity.

Mars Cave Explorer Rover – Thermal Subsystem | Thermal Analysis, Trade Studies, Heat Flow Mapping

Fall 2025

- Collaborated in a 2-person team within a 20-member group to design the thermal subsystem for a Mars cave-exploring rover, operating under a fixed launch mass, limited power budget, and extreme thermal cycling.
- Conducted trade studies to evaluate and select thermal control components (heaters, insulation, thermal interfaces); generated heat flow maps to quantify heat loss rates and determine required heater power to keep electronics within operational limits.
- Developed a thermal risk matrix identifying critical vulnerabilities including overnight heat loss and heater failure; proposed mitigations such as redundancy, insulation layering, and duty-cycle adjustments.
- Presented thermal risks and design trade-offs to the broader mission team, influencing power allocation decisions and overall mission feasibility.